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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Design and Analysis of Algorithms (course)

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Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 :
Introduction
()

Week 1 :
Analysis of
algorithms ()

Week 1 Quiz
()

Week 2 :
Searching
and sorting
()

Week 8 Quiz

Your last recorded submission was on 2025-09-17, 10:03 Due date: 2025-09-17, 23:59 IST.

All questions carry equal weightage. You may submit as many times as you like within the deadline. Your final submission will be graded.

1) Which of the following is a linear constraint?

1 point

- ☐ $8x + 33yz + 14z \geq 21$
☐ $8xy + 33y + 14z \leq 21$
☒ $8x + 33y + 14z = 21$
☐ $8x + 33y + 14xyz \geq 21$

2) The President is arriving to inaugurate a stadium. He will go directly from the airport to the stadium. Security considerations require two routes to be available for the President that do not overlap on any section of road, though the routes can cross each other at intersections. **1 point**

This can be modelled as a network flow problem where the source and target are the airport and the stadium, road intersections are nodes and each road segment is an edge. The actual flow problem to be solved is to:

- ☐ Assign a total of capacity 2 to all outgoing edges from the source and find a feasible flow.
☐ Assign a total of capacity 2 to all incoming edges to the target and find a feasible flow.
☐ Assign each edge capacity 1 and check that the maximum flow is less than 2.
☒ Assign each edge capacity 1 and check that the maximum flow is at least 2.

3) City authorities are concerned about traffic accidents on major roads. They would like to have ambulances stationed at road intersections to quickly reach the scene of any accident along these roads. To minimize response time, ambulances are to be located at intersections with traffic lights so that any segment of road can be reached by at least one ambulance that does not have to pass through a traffic light to reach the scene of the accident. If **1 point**

Week 2 Quiz ()

Week 2 Programmin g Assignment ()

Week 3 : Graphs ()

Week 3 Quiz ()

Week 3 Programmin g Assignment ()

Week 4 : Weighted graphs ()

Week 4 Quiz ()

Week 4 Programmin g Assignment ()

Week 5: Data Structures: Union-Find and Heaps ()

Week 5 : Divide and Conquer ()

Week 5 Quiz ()

Week 6: Data Structures: Search Trees ()

we model the road network as a graph, where intersections with traffic lights are vertices and edges represent road segments between traffic lights, the graph theoretic question to be answered is:

- ☐ Find a spanning tree with minimum cost.
- ☐ Find a minimal colouring.
- ☒ Find a minimum size vertex cover.
- ☐ Find a minimum size independent set.

4) We have an exponential time algorithm for problem A, and problem A reduces in polynomial time to problem B. From this we can conclude that:

1 point

- ☐ B has an exponential time algorithm.
- ☐ B cannot have a polynomial time algorithm.
- ☐ A cannot have a polynomial time algorithm.
- ☒ None of the other choices are correct.

5) Suppose SAT reduces to a problem C. To claim that C is NP-complete, we additionally need to show that:

1 point

- ☒ There is a checking algorithm for C.
- ☐ Every instance of C maps to an instance of SAT.
- ☐ Every instance of SAT maps to an instance of C.
- ☐ C does not have an efficient algorithm.

You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers



**Week 6:
Greedy
Algorithms ()**

**Week 6 Quiz
()**

**Week 6
Programmin
g
Assignment
()**

**Week 7:
Dynamic
Programmin
g ()**

**Week 7 Quiz
()**

**Week 7
Programmin
g
Assignment
()**

**Week 8:
Linear
Programmin
g and
Network
Flows ()**

**Week 8:
Intractability
()**

**Week 8 Quiz
()**

 **Quiz: Week 8
Quiz
(assessment?
name=246)**

**Download
Videos ()**

**Lecture
Material ()**



**Problem
Solving
Session -
July 2025 ()**

